

Multiband Wire Antenna

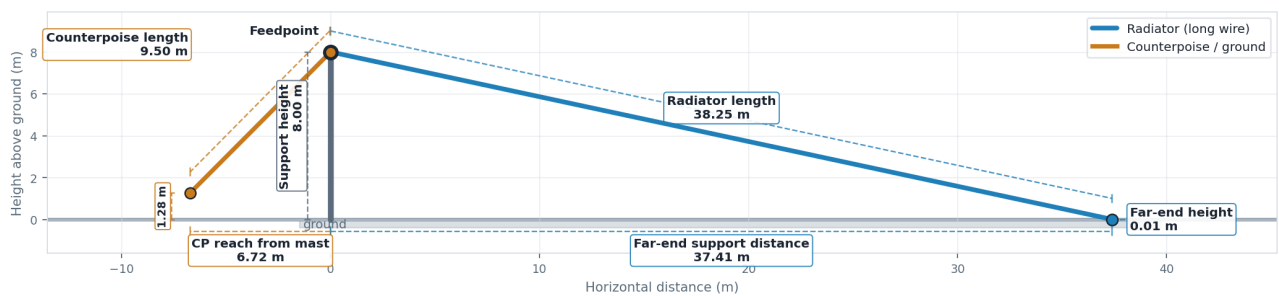
NEC2-Optimized Configuration Datasheet

38.25 m

RADIATOR LENGTH

Configuration Overview

Radiator length	38.250 m
Counterpoise length	9.500 m (45.0° from vertical)
Radiator height	8.00 m
Recommended UnUn ratio	9 : 1
Active bands	40m, 20m, 10m
Optimization mode	NEC2
Combined score (lower = better)	1.220
Wire geometry	Wire geometry : SLOPED (feedpoint z=8.000 m → far end z=0.0100 m)



Performance Report

Per-Band Performance

Band	Freq (MHz)	VSWR	Rating
40m	7.100	2.43	★★★ EXCELLENT
20m	14.175	1.84	★★★ EXCELLENT
10m	28.500	2.15	★★★ EXCELLENT

Per-band impedance (antenna side and transmitter side):

Band	R_ant (Ω)	X_ant (Ω)	R_tx (Ω)	X_tx (Ω)	Source
40m	268.4	+261.8	29.83	+29.08	NEC2
20m	556.4	+290.6	61.82	+32.29	NEC2
10m	485.6	+365.2	53.96	+40.57	NEC2

BEST CANDIDATE — DETAILED BREAKDOWN

Wire length	38.250 m
CP length	9.500 m (45.0° from vertical)
Wire geometry	Wire geometry : SLOPED (feedpoint z=8.000 m → far end z=0.0100 m)
Combined score	1.2203
VSWR penalty	0.4262
Avoidance (active bands)	0.2448
Avoidance (all bands)	0.2448
NEC2 data used	YES

Band	MHz	VSWR	Quality	Avoidance
40m	7.100	2.43 GOOD	★★★ EXCELLENT	0.2500
20m	14.175	1.84 GOOD	★★★ EXCELLENT	0.2343
10m	28.500	2.15 GOOD	★★★ EXCELLENT	0.2500

UnUn RATIO ANALYSIS (for best antenna geometry)

UnUn ratio (this run)	9.0:1
Continuous optimum	8.52:1 (agg. VSWR penalty: 1.3462)
Best standard ratio	9:1 (agg. VSWR penalty: 1.3544)
Recommendation	→ Current ratio 9:1 is already optimal among standard values.

Standard ratio sweep:

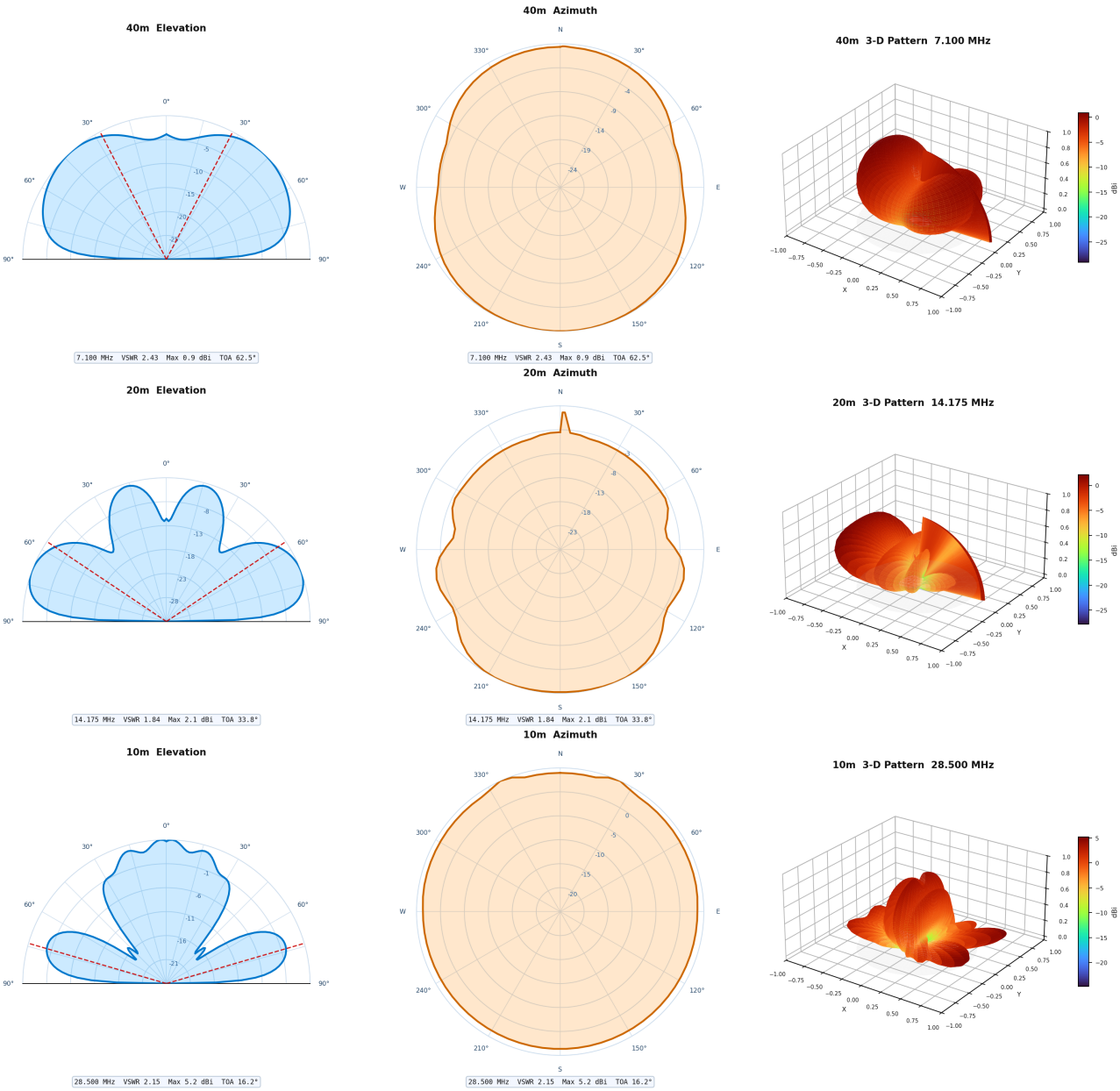
Ratio	Score	VSWR@40m	VSWR@20m	VSWR@10m
1:1	12.2436	10.56	14.18	15.24
1.5:1	10.0640	7.12	9.47	10.19
2:1	8.4778	5.42	7.12	7.68
3:1	5.7400	3.79	4.78	5.18
4:1	3.8201	3.03	3.62	3.96
6:1	2.0160	2.46	2.50	2.79
9:1	1.3544	2.43	1.84	2.15
12:1	1.6657	2.74	1.65	2.00
16:1	2.4381	3.33	1.75	2.13
25:1	4.6132	4.87	2.40	2.83
27:1	5.1247	5.22	2.56	3.01
36:1	6.9605	6.85	3.33	3.87
49:1	8.6888	9.23	4.47	5.16
64:1	10.2927	12.00	5.80	6.68

Antenna-side impedance (independent of UnUn ratio):

Band	R_ant (Ω)	X_ant (Ω)	Z_ant (Ω)	θ (°)	Best ratio
40m	268.4	+261.8	374.9	+44.3	7.50:1
20m	556.4	+290.6	627.7	+27.6	12.55:1
10m	485.6	+365.2	607.6	+36.9	12.15:1

Radiation Patterns

Radiation Diagrams — Wire 38.25 m / CP 9.50 m (45.0°)



Generated automatically by the NEC2 Antenna Length Optimizer (LU3VEA, CC0 v1.0). All values are simulation estimates over average ground; on-site results may vary.